Metallic Cavities

The following subjects will be covered

- Introduction.
- The Metallic Rectangular Cavity.
- TE and TM Modes.
- Examples.

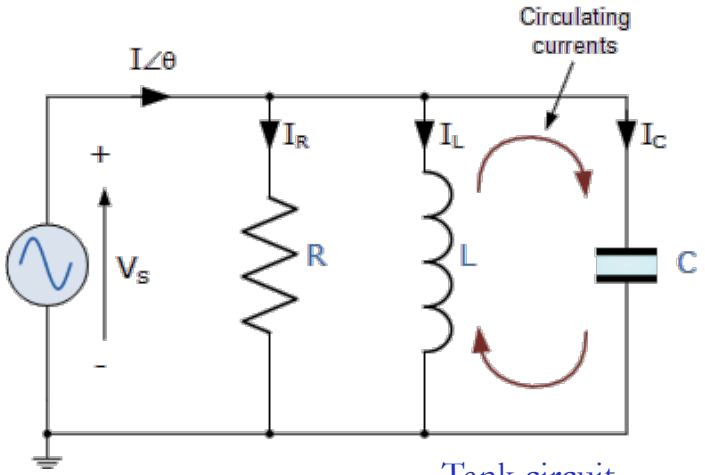
Introduction: Resonators

In electric circuits, a resonant RLC circuit is used to store electric energy. The energy is stored in the magnetic field of the inductor and the electric field of the capacitor. This energy oscillates back and forth between the inductor and the capacitor (tank circuit). This situation makes the currents in the inductor and capacitor always equal and opposite to each other. The energy being drawn from the supply is thus equal to zero (because the vector sum of all three currents mounts to zero). The frequency at which "resonance" happens, f_o, is given below.

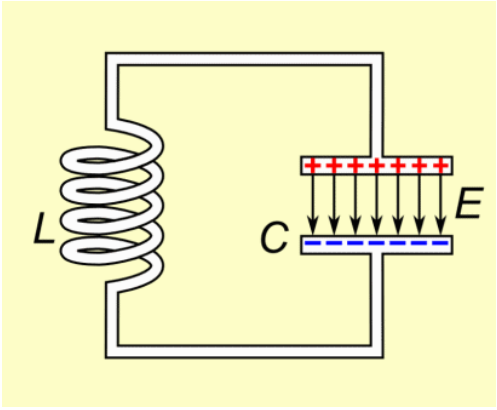
The quality factor, Q, is a measure of the ability of the resonator to retain energy.

$$f_o = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = R\sqrt{\frac{C}{L}}$$

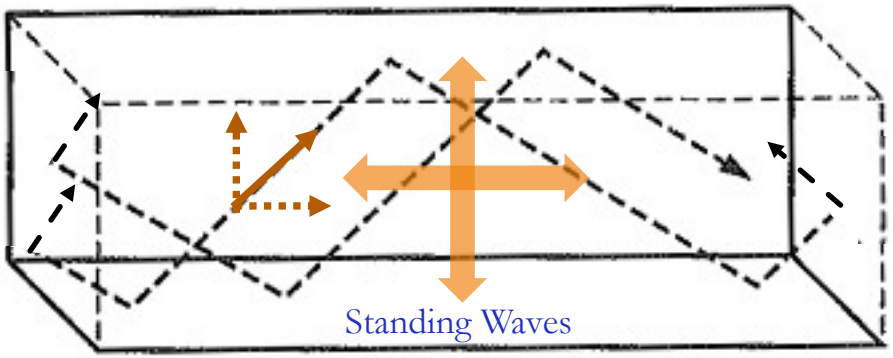


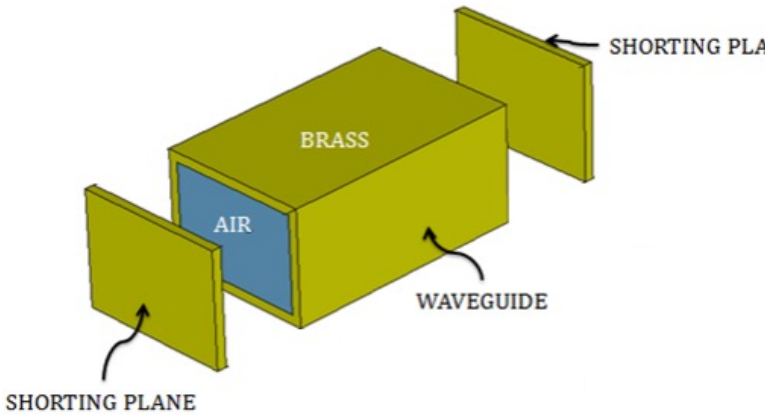
Tank circuit

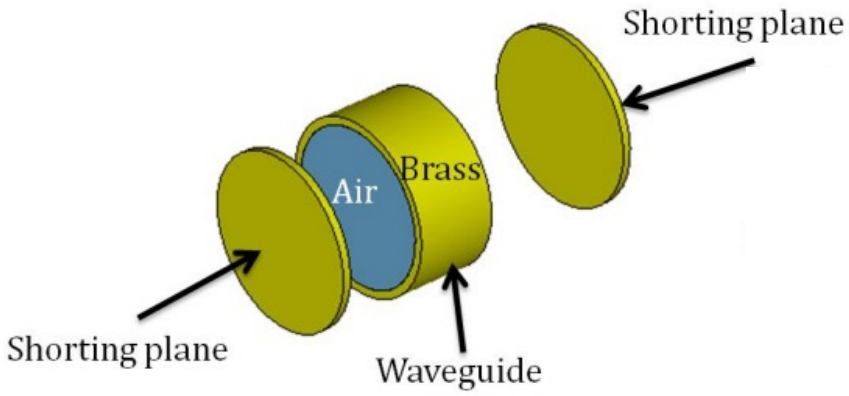


Introduction: Resonators

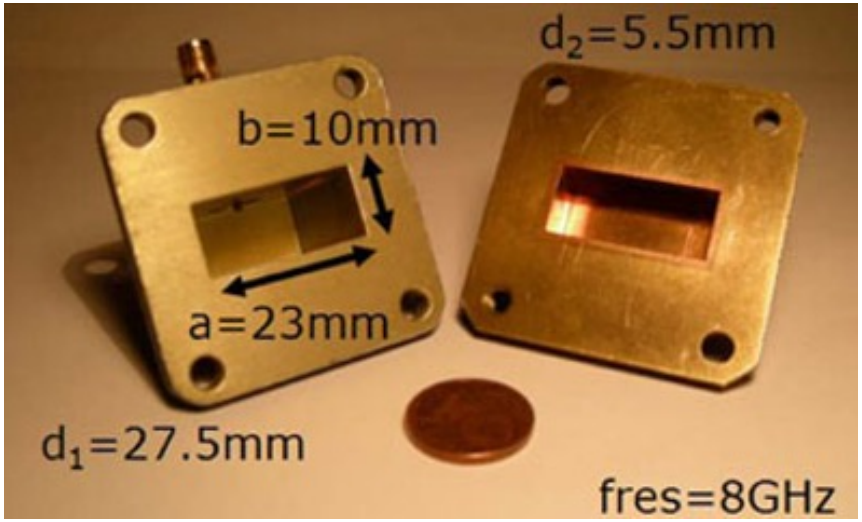
A similar situation can be realized for EM waves, if the electric and magnetic fields are stored inside an enclosure "cavity" made of metal. Because of reflection at metal surfaces, the generation of "standing" waves is possible.

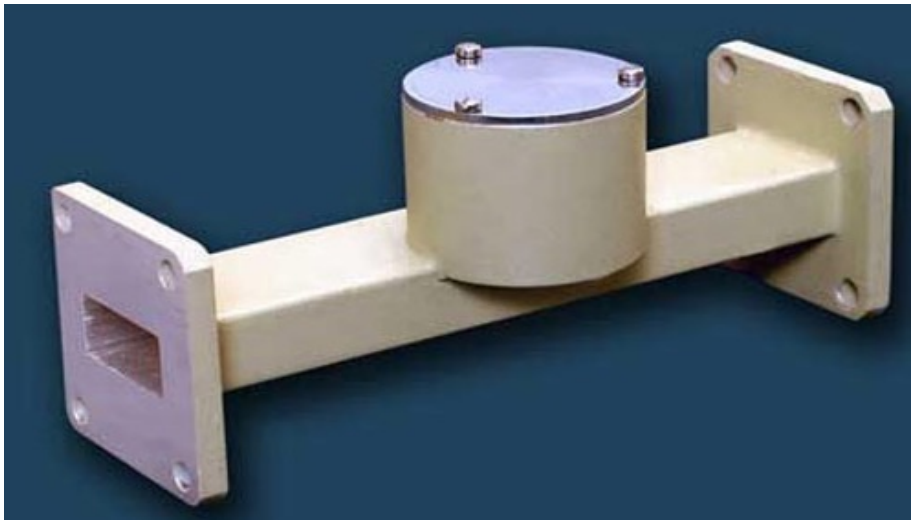






Metallic Cavities (Resonators)











Metallic Cavities (Resonators)

Metallic Cavities are one of the most useful microwave components for EM energy storage and filtering at frequencies above 1 GHz. At frequencies above 1 GHz, normal RLC resonators become impractical and unreliable.

Metallic Cavities provide a low-loss and reliable (high-power handling capability) for EM power applications.

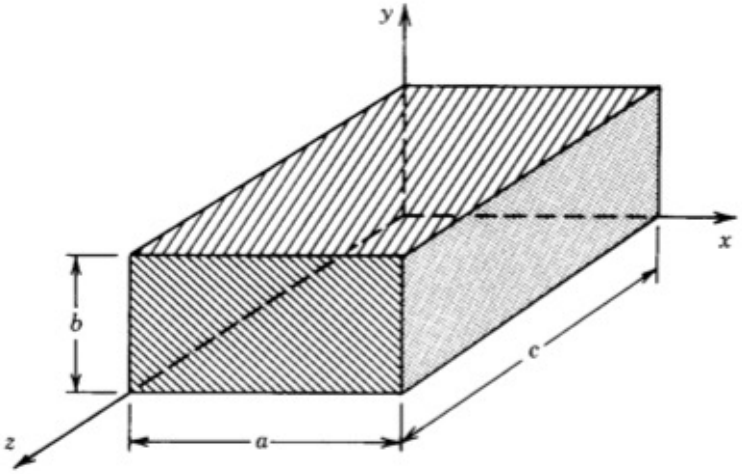
They find applications in

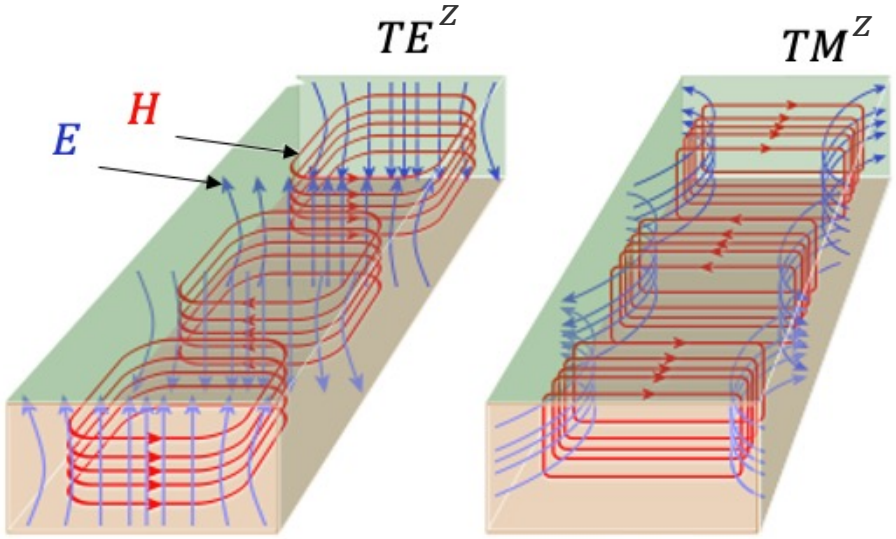
1) Radar equipment.
2) Microwave ovens.
3) Material identification.
4) TV-satellite converters.
5) EM filters.

They come in two basic configurations; rectangular and circular/elliptical.

EM Waves in a Metallic Rectangular Cavity

The concept of wave confinement here is based on perfect reflection over the metallic surfaces. The wave is confined to the space filling the cavity (low-loss).





For a given cavity dimensions (a, b, c), an infinite number of different wave profiles (modes) are possible, each with its own frequency, wavelength, etc.

Waves in MRC are non-TEM; classified as either TE or TM.

Wave Equation Solutions: TM Modes

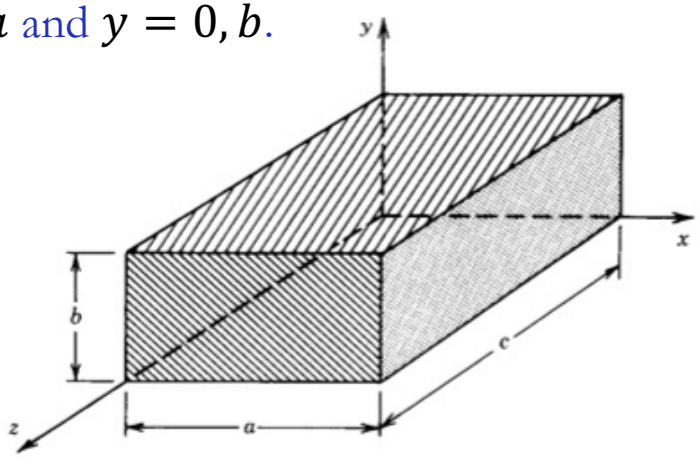
Without going into complicated details, we can construct the required solutions by inspection.

1) TM Modes ($H_z = 0$)

$$E_z = (C_1 \cos\beta_x x + C_2 \sin\beta_x x) \cdot (C_3 \cos\beta_y y + C_4 \sin\beta_y y) \cdot (C_5 \cos\beta_z z + C_6 \sin\beta_z z)$$

Apply boundary conditions, tangential fields=0 at $x = 0, a$ and $y = 0, b$.

$$E_z = E_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot \cos\left(\frac{p\pi}{c}z\right)$$



$m = 1, 2, \cdots$ $\quad n = 1, 2, \cdots$ $\quad p = 0, 1, 2, \cdots$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{b} \qquad \beta_z = \frac{p\pi}{c}$$

$$\beta^2 = \omega^2\varepsilon\mu = \beta_x^2 + \beta_y^2 + \beta_z^2$$

Mode number is TM_{mnp}

Modes TM_{00p} , TM_{01p} and TM_{10p} are not possible ($E_z = 0$)

Wave Equation Solutions: TM Modes

To find other field components, use Maxwell's equations.

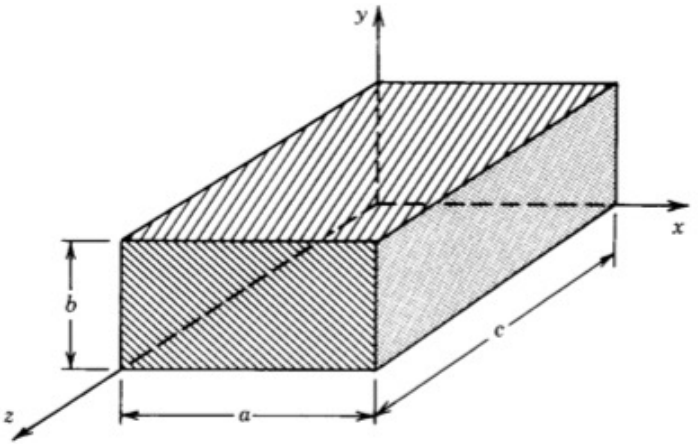
$$E_z = E_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot \cos\left(\frac{p\pi}{c}z\right)$$

$$E_x = \frac{-1}{\beta^2}\left(\frac{m\pi}{a}\right)\left(\frac{p\pi}{c}\right) E_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot \sin\left(\frac{p\pi}{c}z\right)$$

$$E_y = \frac{-1}{\beta^2}\left(\frac{n\pi}{b}\right)\left(\frac{p\pi}{c}\right) E_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot \sin\left(\frac{p\pi}{c}z\right)$$

$$H_x = \frac{j\omega\varepsilon}{\beta^2}\left(\frac{n\pi}{b}\right) E_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot \cos\left(\frac{p\pi}{c}z\right)$$

$$H_y = \frac{-j\omega\varepsilon}{\beta^2}\left(\frac{m\pi}{a}\right) E_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot \cos\left(\frac{p\pi}{c}z\right)$$



$$H_z = 0$$

$$f_{mnp} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

Cut-off (resonance) frequency for mode TM_{mnp}

The dominant mode is the mode that will be excited first and has the lowest resonance frequency (TM_{110}).

Wave Equation Solutions: TE Modes

Without going into complicated details, we can construct the required solutions by inspection.

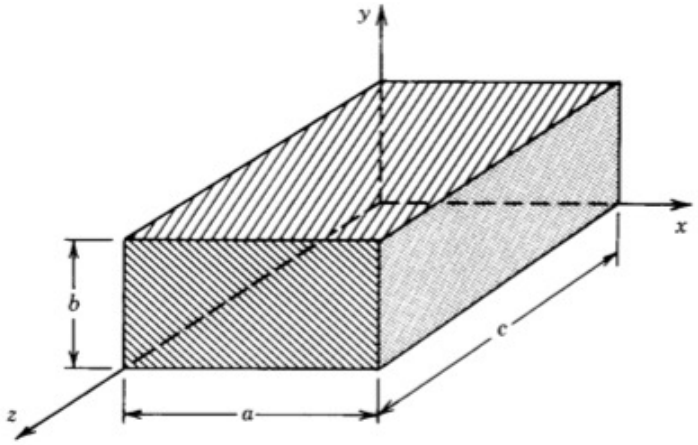
2) TE Modes ($E_z = 0$)

$$H_z = H_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot \sin\left(\frac{p\pi}{c}z\right)$$

$m = 0, 1, 2, \cdots$ $\quad n = 0, 1, 2, \cdots$ $\quad p = 1, 2, \cdots$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{b} \qquad \beta_z = \frac{p\pi}{c}$$

$$\beta^2 = \omega^2\varepsilon\mu = \beta_x^2 + \beta_y^2 + \beta_z^2$$



Modes TE_{mn0} and TE_{00p} are not possible.

Mode number is TE_{mnp}

But modes TE_{m0p}. and TE_{0np} are valid.

Note: m and n cannot be both zero at the same time

Wave Equation Solutions: TE Modes

To find other field components, use Maxwell's equations.

$$H_z = H_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot \sin\left(\frac{p\pi}{c}z\right)$$

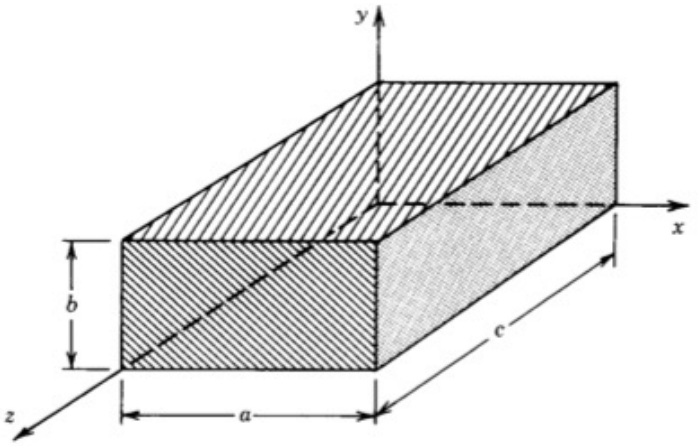
$$H_x = \frac{-1}{\beta^2}\left(\frac{m\pi}{a}\right)\left(\frac{p\pi}{c}\right) H_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot \cos\left(\frac{p\pi}{c}z\right)$$

$$H_y = \frac{-1}{\beta^2}\left(\frac{n\pi}{b}\right)\left(\frac{p\pi}{c}\right) H_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot \cos\left(\frac{p\pi}{c}z\right)$$

$$E_x = \frac{j\omega\mu}{\beta^2}\left(\frac{n\pi}{b}\right) H_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot \sin\left(\frac{p\pi}{c}z\right)$$

$$E_y = \frac{-j\omega\mu}{\beta^2}\left(\frac{m\pi}{a}\right) H_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot \sin\left(\frac{p\pi}{c}z\right)$$

$$E_z = 0$$



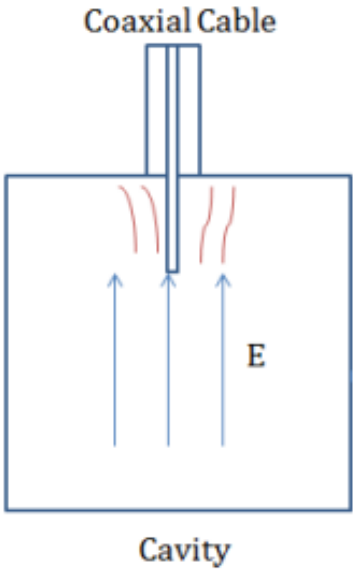
$$f_{mnp} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

Cut-off (resonance) frequency for mode TE_{mnp}

The dominant mode is the mode that will be excited first and has the lowest resonance frequency (TE_{101} or TE_{011}).

Metallic Rectangular Cavity

A cavity can be excited/probed using a number of methods. A coaxial cable tip can be inserted through a small hole, or an aperture can be made in the wall. The probe location has to match the profile of the mode being sensed/excited.



The quality factor for any resonator is defined as,

$$Q = 2\pi \frac{Time\ average\ Energy\ stored}{Energy\ loss\ per\ cycle}$$

For the dominant mode of a metallic rectangular cavity,

$$Q_{TE_{101}} = \frac{abc(a^2 + c^2)}{\delta[2b(a^3 + c^3) + ac(a^2 + c^2)]}$$

where the skin depth of the cavity metallic walls

$$\delta = \frac{1}{\sqrt{\pi f_{101} \mu_o \sigma_c}}$$

σ_c is the metal conductivity

Examples

(1) Consider an MRC with dimensions 5 cm × 4 cm × 10 cm, filled with air and made of copper ($\sigma_c = 5.8\times10^7$ S/m). Find the cutoff frequencies of all modes up to 5 GHz. Find the quality factor of the dominant mode?

$$f_{mnp} = \frac{c}{2}\sqrt{\left(\frac{m}{0.05}\right)^2 + \left(\frac{n}{0.04}\right)^2 + \left(\frac{p}{0.1}\right)^2}$$

4 TE modes
1 TM mode

Dominant mode:
TE_{101}

Single mode operation between 3.35-4.04 GHz

TE_{mnp}	f_{mnp} (GHz)	TM_{mnp}	f_{mnp} (GHz)
011	4.04	110	4.80
101	3.35	111	5.03
111	5.03		
102	4.24		
012	4.80		

$Q_{TE101} = 14{,}358$

Examples

(2) Consider a dielectric-filled rectangular cavity with dimensions a=5 cm, b=3 cm and c=3 cm. One of the TM modes that are excited inside this cavity has x-component magnetic field given by

$$H_x(x,y,z,t) = 36\sin(40\pi x)\cdot\cos(100\pi y)\cdot\sin(6.4\pi\times10^9\, t) \qquad \text{(mA/m)}$$

Find (a) the number of this mode, (b) the dielectric constant of the cavity filling material and (c) all the TE modes that are excited before the given mode.

$$H_x = \frac{j\omega\varepsilon}{\beta^2}\left(\frac{n\pi}{b}\right)E_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot\cos\left(\frac{p\pi}{c}z\right)$$

$$\beta_x = \frac{m\pi}{a} = 40\pi \;\rightarrow\; m = 2$$

$$\beta_y = \frac{n\pi}{b} = 100\pi \;\rightarrow\; n = 3$$

$$\beta_z = \frac{p\pi}{b} = 0 \;\rightarrow\; p = 0$$

TM_{230}

$$f_{230} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{\left(\frac{2}{0.05}\right)^2+\left(\frac{3}{0.03}\right)^2+\left(\frac{0}{0.03}\right)^2} = 3.2\times10^9 \text{ (Hz)}$$

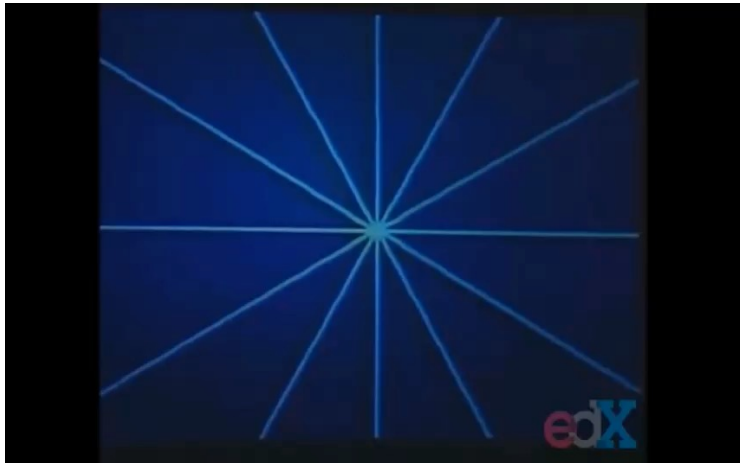
$\varepsilon_r = 25$

TE Modes

f_{110} = ✗ 1.166e+9	f_{101} = ✓ 1.166e+9	f_{011} = ✓ 1.414e+9
f_{111} = ✓ 1.536e+9	f_{201} = ✓ 1.562e+9	f_{120} = ✗ 2.088e+9
f_{211} = ✓ 1.855e+9	f_{210} = ✗ 1.562e+9	f_{021} = ✓ 2.236e+9
f_{220} = ✗ 2.332e+9	f_{221} = ✓ 2.538e+9	f_{121} = ✓ 2.315e+9

Antennas

The following subjects will be covered



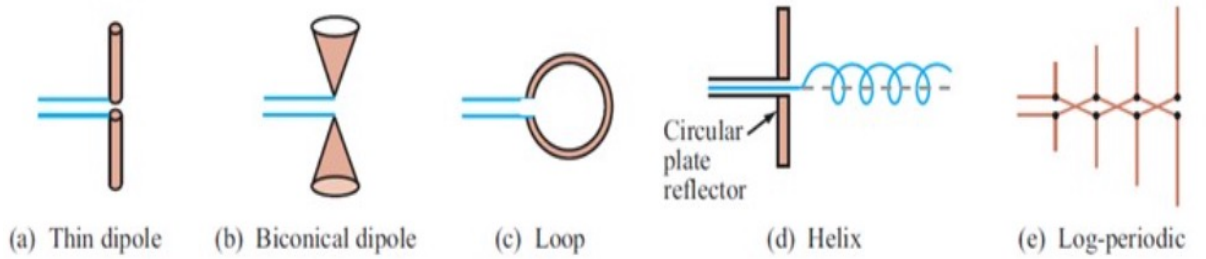
- Introduction.
- The Hertzian (point) Antenna
- The Dipole Antenna
- The Monopole Antenna.
- The Microstrip (patch) Antenna

Antennas: Introduction

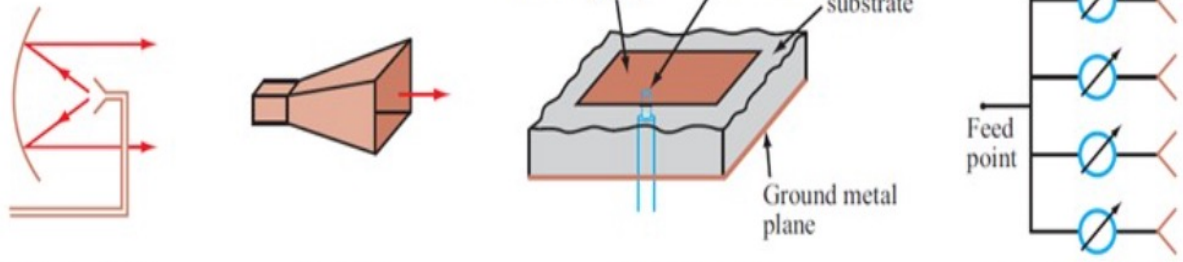
Antennas are metallic structures that are designed specifically to radiate (receive) EM waves efficiently, within a specific frequency band, with specific characteristics.

They come in many different types and shapes, each with

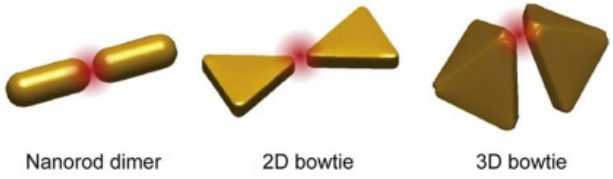
- Basic shapes.
- Principle of radiation.
- Field profiles (modes).
- Frequency range.
- Radiation patterns.
- Advantages.
- Limitations.



(a) Thin dipole (b) Biconical dipole (c) Loop (d) Helix (e) Log-periodic



(f) Parabolic dish reflector (g) Horn (h) Microstrip (i) Antenna array



Nanorod dimer 2D bowtie 3D bowtie

Antennas: Introduction

Antennas are characterized based on several parameters. The application of a certain type of antenna in a certain application is based on the nature of the problem.

Basic parameters include:

1) Resonance frequency: the frequency at which the antenna efficiently converts incoming power to EM waves, or have best reception.
2) Bandwidth: the range of frequencies where the operation of an antenna is acceptable.
3) Polarization: linear, circular or elliptical.
4) Radiation pattern: the space map of antenna radiation.
5) Efficiency: the conversion ratio of supplied power to radiated power.
6) Gain: the ratio of radiated power in a certain direction to a reference antenna.
7) Directivity: a measure of the focus of a radiation beam.

The Hertzian Dipole

This is a basic antenna element, a very short wire carrying current, or a differential antenna (a point source).

Basic assumptions:

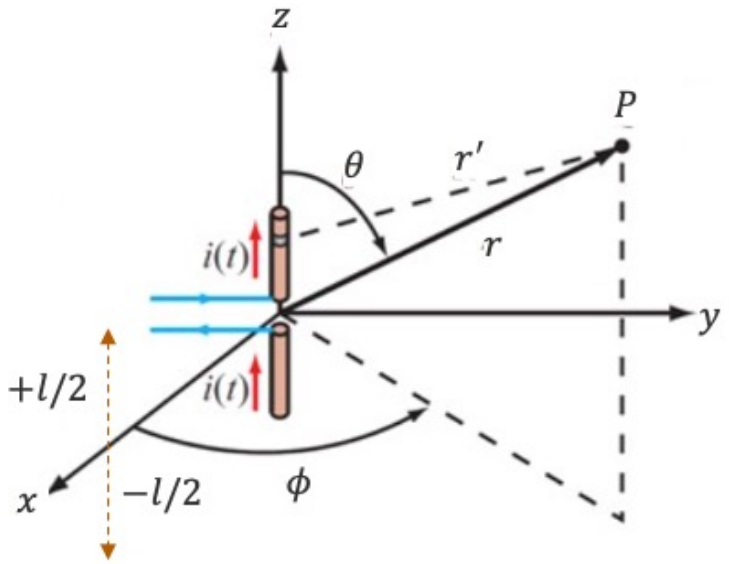
1) Infinitesimal length compared to distances involved (the wavelength)

$$l \ll \lambda \qquad l < \lambda/50$$

2) The current along the wire becomes constant.

$$I(z) = I_o \cos \omega t \approx I_o$$

3) The radiated waves are spherical.



r: Range ϕ: Azimuthal θ: Zenith

Spherical coordinates simplify the work

$$\vec{E} = E_r \vec{a}_r + E_\phi \vec{a}_\phi + E_\theta \vec{a}_\theta$$

$$\vec{H} = H_r \vec{a}_r + H_\phi \vec{a}_\phi + H_\theta \vec{a}_\theta$$

Because the current flows in the z-direction, the resulting fields have the following components

$$E_r, E_\theta, H_\phi$$

$$E_\phi = H_\theta = H_r = 0$$

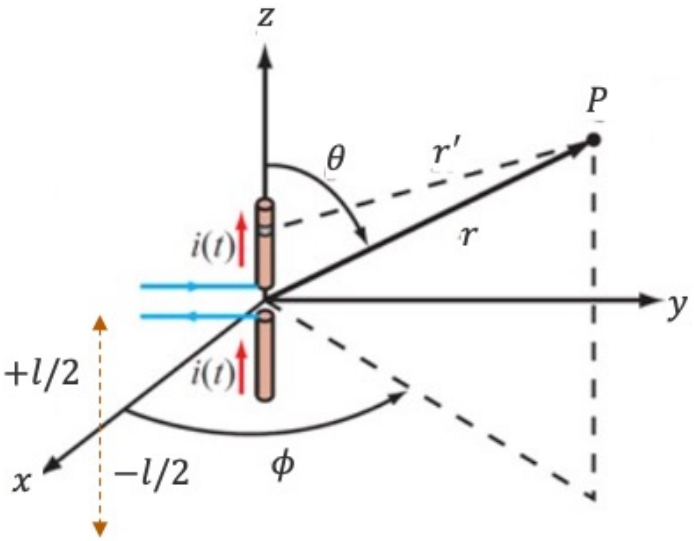
The Hertzian Dipole

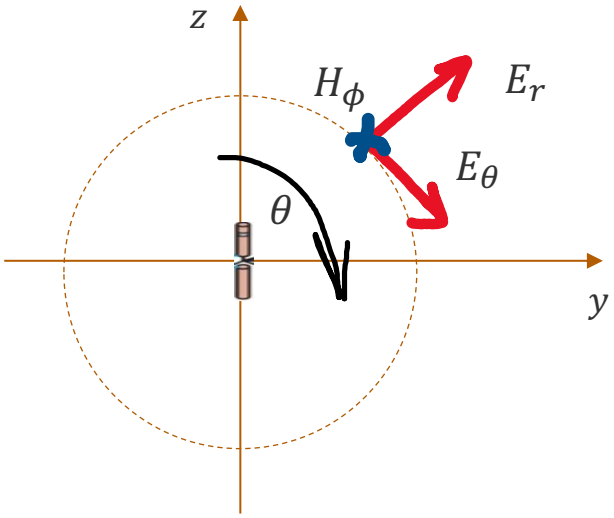
The existing field are as follows.

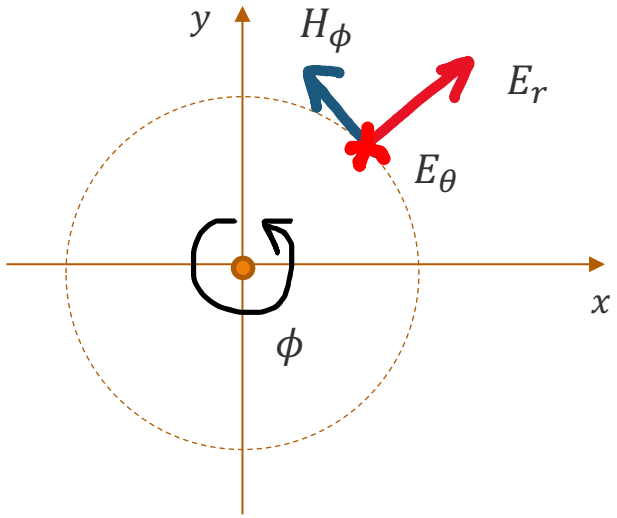
$$E_r = \frac{2I_o\, l\, \beta^2}{4\pi}\eta_o \left[\frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3}\right]\cos\theta\; e^{-j\beta r}$$

$$E_\theta = \frac{I_o\, l\, \beta^2}{4\pi}\eta_o \left[\frac{j}{\beta r} + \frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3}\right]\sin\theta\; e^{-j\beta r}$$

$$H_\phi = \frac{I_o\, l\, \beta^2}{4\pi}\left[\frac{j}{\beta r} + \frac{1}{(\beta r)^2}\right]\sin\theta\; e^{-j\beta r}$$







The Hertzian Dipole

Far-field Approximation.

r very large $\quad \beta r \gg 1$

$$\frac{2\pi}{\lambda} r \gg 1 \rightarrow r \gg \frac{\lambda}{2\pi}$$ Far-field Zone

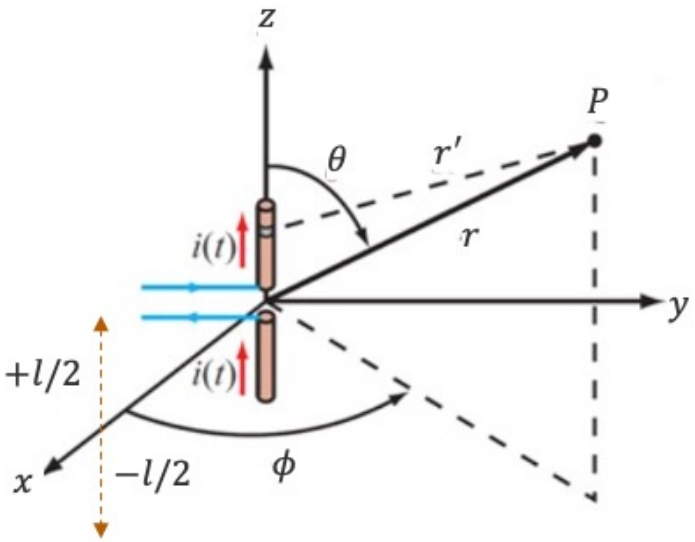
$$E_r \approx 0$$

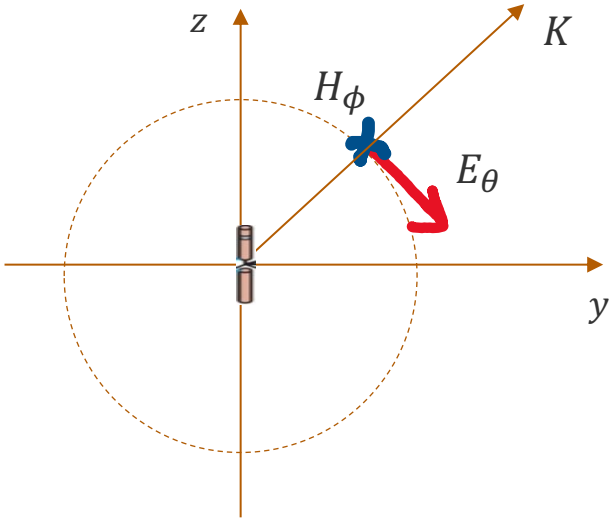
$$E_\theta = \frac{jI_o\, l\, \beta}{4\pi} \eta_o \sin\theta \left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = \frac{jI_o\, l\, \beta}{4\pi} \sin\theta \left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = E_\theta / \eta_o$$

Plane-wave behavior





The Hertzian Dipole

Time-average Power density

$$\vec{S}_{ave} = \frac{1}{2} Re(\vec{E} \times \vec{H}^*)$$

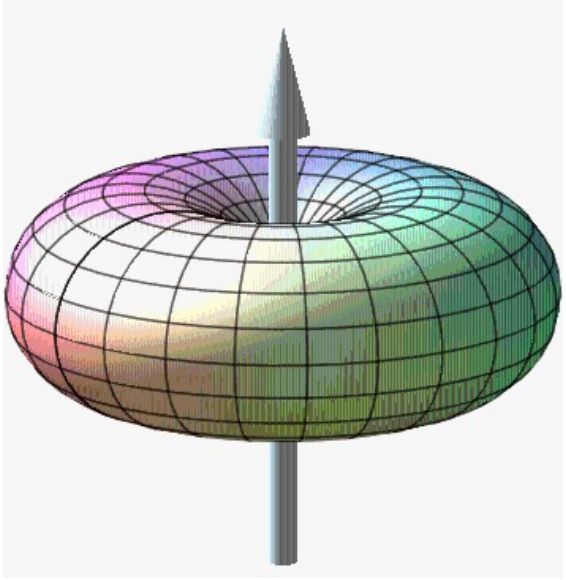
$$\vec{S}_{ave} = \frac{\eta_o \beta^2 I_o^2 l^2}{32 \pi^2 r^2} (\sin\theta)^2 \ \vec{a}_r$$

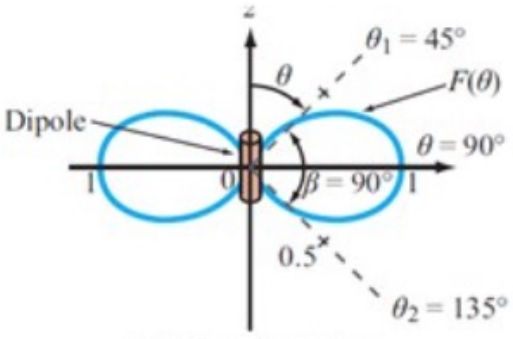
$$\vec{S}_{ave} = \frac{\eta_o I_o^2}{8r^2} \left(\frac{l}{\lambda}\right)^2 (\sin\theta)^2 \ \vec{a}_r$$

$$\vec{S}_{ave} = S_o(r) \ (\sin\theta)^2 \ \vec{a}_r$$

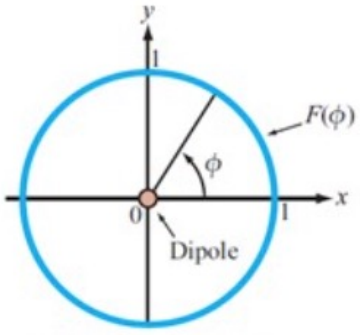
$$S_o(r) = \frac{\eta_o I_o^2}{8r^2} \left(\frac{l}{\lambda}\right)^2$$

$$S_{ave_max} = S_o(r)$$





Elevation plane $\phi = 0$



Azimuthal plane $\theta = \frac{\pi}{2}$

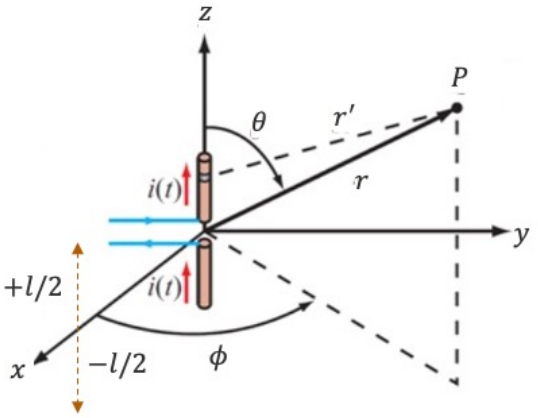
The Hertzian Dipole

Radiated Power

$$P_{rad} = \oint_S \vec{S}_{ave} \cdot \vec{d}_S$$

$$P_{rad} = \int_0^{2\pi} \int_0^{\pi} \frac{\eta_o I_o^2}{8r^2} \left(\frac{l}{\lambda}\right)^2 (\sin\theta)^2 r^2 \sin\theta \, d\theta d\phi$$

$$P_{rad} = \frac{\pi \, \eta_o I_o^2}{3} \left(\frac{l}{\lambda}\right)^2$$



Radiation Resistance

$$P_{rad} = I_{rms}^2 \, R_{rad}$$

$$P_{rad} = \frac{1}{2} I_o^2 \, R_{rad}$$

$$R_{rad} = 80\,\pi^2 \left(\frac{l}{\lambda}\right)^2 \approx 1\ \Omega$$

Example

The maximum signal picked from a radio station 2 km away has a magnetic field magnitude of 5 μA/m. Find the station radiated power. $l/\lambda = 1/25$.

$$|H_\phi| = \frac{I_o \, l \, \beta}{4\pi r} \sin\theta \ = 5\mu \rightarrow I_o = 0.5\ A$$

$$P_{rad} = \frac{\pi \, \eta_o I_o^2}{3} \left(\frac{l}{\lambda}\right)^2 = 157.7\ mW$$

$$R_{rad} = 80\,\pi^2 \left(\frac{l}{\lambda}\right)^2 = 1.26\ \Omega$$

The Hertzian Dipole

Efficiency

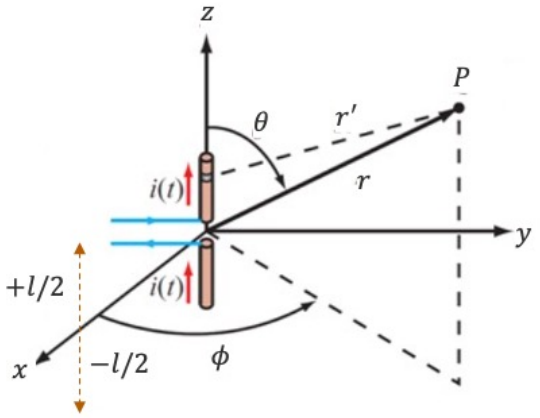
$$\eta = \frac{P_{rad}}{P_{tot}} = \frac{P_{rad}}{P_{rad} + P_{loss}}$$

Gain

$$G = \frac{S_o}{P_{tot}/4\pi r^2}$$

Directivity

$$D = \frac{S_o}{P_{rad}/4\pi r^2} \approx 1.5$$



Example: A 1-m dipole is excited by a 5-MHz current of amplitude 5 A. Find the broadside power density at 2 km.

Check if it's a Hertzian dipole. $\lambda = \frac{v}{f} = 3\times10^8/5\times10^6 = 60\ m$ $\qquad \frac{l}{\lambda} = \frac{1}{60}$

$$S_{ave} = \frac{\eta_o I_0^2}{8(2000)^2}\left(\frac{l}{\lambda}\right)^2 (\sin 90°)^2 = 8.2\times10^{-8}\ \text{W/m}^2$$

Q & A

Send me an email if you have any questions.